INCH-POUND

A-A-52035
December 13, 1991

COMMERCIAL ITEM DESCRIPTION

PROTRACTOR, ONE ARM: SEMICIRCULAR

(CORROSION-RESISTANT ALLOY, WITH CASE)

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-P-3756.

This description covers levels used for making differential elevation measurements in surveying and in other engineering applications.

Salient characteristics:

Protractor. The protractor shall consist of a semicircular arc 8 inches in diameter, a 6-inch adjustable vernier arm, and a 1/2 vernier reading to one minute.

Material. The protractor shall be of a corrosion resistant nickel silver material.

Graduations and numbering. The graduations and numbering shall be cut into the surface of the protractor on the outer circular edge, and shall be machine divided. Graduation lines shall be 0.005 inch ±0.001 inch wide at +72 degrees Fahrenheit, and all graduation lines shall be within 0.0012 inch of their true relative position. All graduation marks shall be V-shaped, uniform, and 0.005 inch deep. The arc of graduation lines shall extend at least 15 degrees below the horizontal centerline.

Centerpiece. The centerpiece of the protractor shall be plastic, not less than 0.060 inch thick and 1/2 inch in diameter.1

Arm. The arm shall pivot about the center of the protractor and shall extend at least 6 inches beyond its outer edge. The arm shall be provided with a double vernier graduated to read directly to 1 minute

Beneficial comments recommendations, additions, clarifications, deletions, etc., and any data which may be of use in improving this document should be sent to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

AMSC N/A FSC 6675

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of arc. Width and tolerance of the graduation marks on the vernier shall exactly match the graduation marks on the protractor. A small knob shall be placed near the tip of the arm to provide a conventional means of rotating the arm.

Clamp and tangent screw. A clamp and tangent screw shall be provided. The clamp shall be activated by a thumbscrew and shall hold the arm in any set position throughout the working range of the protractor. When released, the clamp shall slide freely with the arm. The tangent screw shall provide a positive and precise means of setting the vernier. The clamp lock and tangent screws shall be the same material as the protractor.

Finish. All metal, and plastic surfaces, except screw threads and knurled edges, shall have a smooth satin finish. Graduations, numbers, and marking shall be filled with permanent black paint or ink.

Case. The protractor shall be provided with a fitted plastic case or a wooden case, with hinged cover. The case shall be lined with a soft fungus-resistant material to protect the instrument during storage and transit.

Fasteners and hinges. Fasteners and hinges shall be corrosion-resistant metal.

Workmanship. All parts of the protractor shall be clean and free of burrs and sharp edges.

Contractor certification. The Contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of the Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

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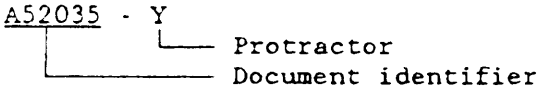
The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. In accordance with Federal Acquisition Regulation, section 23.403, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

Part or identification number (PIN). The PIN to be used for protractors acquired to this commercial item description is created as follows:



Note:
1. Title, number, and date of this item description.
2. When this commercial item description is used for procurement, the commercial item certification clause must appear in the solicitation.
3. FED-STD-376 is available from the Naval Publishing and Printing Service Office (NPPSO), Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.
4. The preparing activity for this item description is the US Army Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Ft. Belvoir, VA 22060-5606.

MILITARY INTERESTS:

Custodians:
Army - ME
Air Force - 99

Review Activity:
DLA-GS

User Activities:
Navy - MC, YD

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
VA-OSS
COM-NBS

PREPARING ACTIVITY:
Army - ME

Project 6675-0944

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